Description

Image





Caption

1. Cervical cage made from PEEK with a modulus very close to that of bone and excellent toughness and fatigue resistance. © Eisertech, LLC 2. Complex PEEK injection part for Oil & Gas and automotive industry. © MING-LI Precision Steel Molds CO.

The material

Polyetheretherketone (PEEK) is a high-performance thermoplastic, meaning that - among thermoplastics - it has exceptionally high stiffness, strength and resistance to heat. This comes at a price: PEEK is 50 times more expensive than PP, and 10 to 20 times more than nylon. This limits it use to applications in which technical performance is paramount.

Compositional summary

(O-(C6H4)-O-(C6H4)-C(O)-(C6H4))n

General properties

Density	1.3e3	-	1.32e3	kg/m^3
Price	* 46.1	-	47.9	GBP/kg
Date first used	1975			

Mechanical properties

Young's modulus	* 3.76	-	3.95	GPa
Shear modulus	* 1.36	-	1.43	GPa
Bulk modulus	* 5.48	-	5.75	GPa
Poisson's ratio	0.39	-	0.41	
Yield strength (elastic limit)	90	-	110	MPa
Tensile strength	97	-	117	MPa
Compressive strength	* 111	-	141	MPa

Elongation		30	-	150	% strain
Hardness - Vickers	*	28	-	31	HV
Fatigue strength at 10^7 cycles	*	39	-	47	MPa
Fracture toughness	*	2.73	-	4.3	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.0101	-	0.0106	

Thermal properties

Melting point		322	-	346	°C
Glass temperature		143	-	157	°C
Maximum service temperature		239	-	260	°C
Minimum service temperature	*	-70.2	-	-50.2	°C
Thermal conductor or insulator?		Good insulator			
Thermal conductivity	*	0.24	-	0.26	W/m.°C
Specific heat capacity		1.34e3			J/kg.°C
Thermal expansion coefficient		40	-	55	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity		3.3e21	-	3e22	µohm.cm
Dielectric constant (relative permittivity)		3.1	-	3.3	
Dissipation factor (dielectric loss tangent)	*	0.0015	-	0.0017	
Dielectric strength (dielectric breakdown)	*	16.7	-	20	MV/m

Optical properties

Transparency	Opaque

Critical Materials Risk

High critical material risk?	No

Processability

Castability	1	-	2	
Moldability	4	-	5	
Machinability	3	-	4	
Weldability	5			

Durability: water and aqueous solutions

Water (fresh)	Excellent
Water (salt)	Excellent
Soils, acidic (peat)	Excellent
Soils, alkaline (clay)	Excellent

Wine	Excellent

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Excellent
Citric acid (10%)	Acceptable
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Excellent
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Excellent
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Excellent
Sulfuric acid (10%)	Excellent
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Acceptable
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Acceptable
Vegetable oils (general)	Excellent
White spirit	Acceptable

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Acceptable
Acetone	Excellent

Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Excellent
Fluorine (gas)	Limited use
O2 (oxygen gas)	Limited use
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Self-extinguishing

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Excellent
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 15.9	-	17.5	kg/kg
Embodied energy (virgin grade production)	* 286	-	315	MJ/kg
Water usage	* 534	-	1.6e3	l/kg

Material processing: energy

Polymer extrusion, embodied energy	* 8.39	-	9.3	MJ/kg
Polymer molding (injection, blow), embodied energy	* 30.3	-	34.7	MJ/kg
Machining, coarse, embodied energy	* 19.6	-	23	MJ/kg
Machining, fine, embodied energy	* 29.6	-	36.7	MJ/kg
Machining, grinding, embodied energy	* 39.5	-	50.5	MJ/kg

Material processing: climate change

Polymer extrusion, climate change (CO2-eq)	* 0.558	-	0.62	kg/kg
Polymer molding (injection, blow), climate change (CO2-eq)	* 2.23	-	2.56	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.28	-	1.51	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.93	-	2.41	kg/kg
Machining, grinding, climate change (CO2-eq)	* 2.59	-	3.31	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 5.25	-	5.78	kg/kg
Embodied energy, recycling	* 54.3	-	59.9	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 30.5	-	32	MJ/kg
Combustion CO2	* 2.83	-	2.97	kg/kg
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Peek can be recycled if unreinforced and uncontaminated.

Recycle mark



Supporting information

Design guidelines

PEEK can be used up to temperatures of 300 C for a short time and 250 C for a long time. It offers high hardness and therefore abrasion resistance; it has excellent fatigue properties and good creep resistance. It has a low coefficient of friction, a low flammability, and low smoke emission during combustion. Chemical resistance is very good (and retained to the same high temperatures) and there is very low water absorption. Unreinforced PEEK offers the highest elongation and toughness of all PEEK grades. Glass-reinforcement significantly reduces the expansion rate and increases the flexural modulus. PEEK can be used as a matrix in continuous carbon fiber composites. Carbon-reinforced PEEK has high compressive strength and stiffness and low expansion coefficient, and its thermal conductivity can be 3 times better than pure PEEK. Processing PEEK is not difficult, despite its high heat resistance, provided the temperature is held at 375 C. It can be injection molded, extruded (into rod, profile, film or wire insulation) and compression molded. It is available as extruded film and sheet in thicknesses from approximately 0.001" to 0.040".

Technical notes

PEEK is a semi-crystalline thermoplastic. It has a high glass transition temperature (Tg = 150 C) and can be used well above this temperature, but its stiffness falls and its expansion coefficient rises above Tg.

Typical uses

Electrical connectors, hot water meters, F1 engine components, valve and bearing components, wire and cable coatings, film and filament for specialized applications, pump wear rings, electrical housing, bushings, bearings.

Tradenames

Ketron PEEK, Thermocomp, Victrex

Links

ProcessUniverse

Producers

Reference